

PANAMA PACIFICO, Panama

WMO: 787955

Lat: 8.917N

Lon: 79.600W

Elev: 43

StdP: 14.67

Time Zone: -5.00 (W05)

Period: 90-99

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10	71.9	73.2	64.4	90.7	88.7	66.1	96.2	86.7	22.5	81.4	20.8	81.0	2.8	10	0.371

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	15.8	95.0	77.8	93.5	77.5	92.9	77.4	81.8	89.1	81.0	88.2	80.4	87.5	11.8	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
80.0	155.7	86.8	79.0	151.0	85.9	78.7	149.2	85.5	45.7	89.9	44.8	89.1	44.2	88.2	87.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.1	16.1	14.3	69.2	97.8	1.3	1.3	68.3	98.7	67.5	99.5	66.8	100.2	65.9	101.2		
(5)			67.1	85.0	1.1	1.4	66.3	86.0	65.6	86.8	65.0	87.5	64.2	88.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	82.6	82.8	83.8	84.6	84.8	83.4	82.4	82.5	81.9	81.5	80.8	80.9	81.7
	DBStd	2.34	1.81	1.74	1.94	2.05	2.29	2.17	2.30	2.00	2.02	1.76	1.61	1.74
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	11893	1018	945	1071	1043	1037	973	1007	988	944	955	928	984
	CDD65	6418	553	525	606	593	572	523	542	523	494	490	478	519
	CDH74	63713	5431	5464	6690	6637	5979	5313	5419	5048	4577	4239	4109	4807
	CDH80	22384	2074	2349	3040	2921	2123	1700	1698	1474	1286	1109	1031	1579
Wind	WSAvg	6.7	8.1	9.3	9.5	8.2	6.2	5.2	6.0	5.7	5.3	5.6	5.4	6.4
Precipitation	PrecAvg	70.8	2.0	0.7	0.4	3.4	7.4	8.1	6.5	7.9	7.5	11.2	10.8	5.0
	PrecMax	101.8	10.4	3.8	2.0	8.5	13.4	17.0	11.4	12.9	13.0	24.3	26.0	10.8
	PrecMin	50.7	0.0	0.0	0.0	0.3	4.5	3.2	2.7	3.5	3.4	5.3	5.0	0.8
	PrecStd	14.1	2.8	0.9	0.6	2.7	2.7	3.6	2.5	2.7	2.4	4.9	4.6	2.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	93.5	95.1	95.4	96.9	95.1	92.9	92.9	91.5	91.5	90.0	89.8	91.8
		MCWB	77.9	77.6	77.1	76.9	80.0	81.4	80.7	80.4	80.1	79.9	78.8	79.0
	2%	DB	91.8	93.4	94.9	95.0	93.0	90.0	91.1	89.7	89.6	88.2	88.1	90.0
		MCWB	77.0	76.3	77.0	77.8	80.0	80.2	80.2	79.9	79.7	79.5	78.7	77.7
	5%	DB	91.1	91.8	93.3	93.3	91.1	89.3	89.4	88.1	87.9	87.4	87.4	89.4
		MCWB	76.7	75.8	76.5	77.5	79.4	79.9	79.7	79.8	79.6	79.3	78.6	77.4
	10%	DB	89.5	91.2	91.7	91.6	89.2	87.6	87.7	86.4	86.1	85.8	85.8	87.7
		MCWB	76.3	75.7	76.2	77.3	79.2	79.7	79.6	79.0	78.8	78.5	78.2	77.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	81.0	80.3	81.0	82.5	82.7	82.6	82.5	81.7	81.4	81.2	80.8	81.3
		MCDB	89.4	91.6	92.3	90.8	90.9	89.3	89.2	88.8	88.3	87.1	87.1	89.0
	2%	WB	79.9	79.1	79.4	80.2	81.0	81.3	81.1	80.5	80.2	80.0	79.7	79.8
		MCDB	88.1	89.9	90.8	89.5	88.4	87.8	88.3	87.4	86.8	85.9	86.1	87.3
	5%	WB	78.8	78.0	78.3	79.2	80.1	80.3	80.1	79.9	79.7	79.4	79.0	78.9
		MCDB	86.5	88.0	89.2	88.5	87.0	86.5	86.9	86.2	85.9	85.1	85.1	86.3
	10%	WB	77.6	77.1	77.3	78.5	79.6	79.7	79.5	79.3	79.0	78.8	78.2	78.1
		MCDB	85.3	86.3	87.7	87.6	86.1	85.4	85.7	85.1	84.6	84.3	83.8	85.0
Mean Daily Temperature Range	5% DB	MDBR	14.3	16.1	16.9	15.8	12.4	11.3	11.3	11.2	11.2	11.1	10.8	12.8
		MCDBR	15.8	17.3	17.8	17.5	14.7	13.1	13.1	12.8	12.8	13.0	12.4	14.6
		MCWBR	5.3	5.3	5.4	5.6	5.6	6.0	6.1	6.1	6.1	6.3	5.7	6.0
	5% WB	MCDBR	13.6	15.3	16.1	15.1	12.7	11.9	11.9	11.7	11.8	11.8	11.2	12.9
		MCWBR	5.9	5.7	5.9	6.0	5.7	6.0	6.2	6.0	6.0	6.2	5.8	6.0
Clear-Sky Solar Irradiance	taub	0.422	0.451	0.465	0.479	0.469	0.445	0.456	0.459	0.429	0.414	0.433	0.425	
	taud	2.413	2.319	2.283	2.250	2.315	2.386	2.360	2.347	2.424	2.493	2.427	2.434	
	Ebn at Noon	282	277	273	266	263	266	265	268	279	284	277	279	
	Edh at Noon	37	42	44	45	42	38	39	41	38	35	37	36	
All-Sky Solar Radiation	RadAvg	1654	1789	1803	1745	1460	1334	1319	1362	1399	1368	1276	1439	
	RadStd	77	82	91	76	108	85	84	73	80	65	102	146	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	N/A	N/A	-0.67	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page